

The X/P Asymmetry Was Never a Law: A Four-Way Split at Four Grade Levels

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Question. A companion note shows that among generalized Inönü–Wigner contractions of the imaginary octonions sharing the Günaydin–Lüst–Malek $1+3+3$ grading shape, genuine (non-associative) R-flux arises if and only if the momentum subspace, not the coordinate subspace, is the one that closes into a subalgebra on its own — an exact, exhaustively verified asymmetry, but explicitly scoped to that one shape. Does this asymmetry reflect something structural about which side of a grading is "allowed" to carry genuine content, or was it an artifact of that shape specifically having no room for the other side's self-bracket to ever survive?

Answer. The latter. Two natural 4-level shapes turn out to be combinatorially impossible outright — one provably, in general, by a clean Fano-plane argument. A third does exist, and once tested exhaustively, the clean asymmetry dissolves into a four-way split: either side can be the one required to stay open, or both can carry genuine content simultaneously. The original rule was never a law about coordinates versus momenta — it was a direct consequence of the $1+3+3$ shape leaving nowhere for the momentum self-bracket to land, so its closure was free regardless of any deeper principle.

1. Background

The companion result: in the $1+3+3$, grade- $(1, 2, 3)$ family, $\{X, X\}$ (grade $1+1=2$) has a real target — grade 2, occupied by P — while $\{P, P\}$ (grade $2+2=4$) does not, since no grade-4 level exists in a 3-level system. Consequently P 's closure never costs anything: the bracket it would kill was already forced to vanish by the shape itself. This section tests whether the same asymmetry survives once a shape actually provides both self-brackets somewhere to land.

2. Two shapes, ruled out first

Attempt 1: grades 1, 2, 3, 4, sizes 3, 2, 1, 1 (X, P, Q, W). Exhaustive search over all 420 possible assignments of the seven octonion generators to this shape found zero satisfying the Weimar–Woods validity condition. The mechanism was traced by direct inspection for representative cases (every failure involved an X - P pair, grade 3, landing on W , grade 4) but no fully general combinatorial proof of impossibility was derived for this specific shape; it is reported here as a verified computational fact.

Attempt 2: grades 1, 2, 4 (skipping 3), sizes 3, 3, 1 (X, P, W). Also exhaustively found impossible — 140 assignments, zero valid. Here a clean, general proof is available.

Lemma (Fano matching obstruction). For any point w among the seven imaginary octonion units, the three unordered pairs $\{a, b\}$ satisfying $a \times b = \pm w$ partition the remaining six points into a perfect matching.

Proof. w lies on exactly three of the Fano plane's seven lines. Each such line contains w and two other points, whose product is $\pm w$. Any two distinct lines of the Fano plane meet in exactly one point; since both lines here pass through w , they share no other point. So the "other two points" on each of the three lines through w are pairwise disjoint, giving three disjoint pairs — six points total, exactly the six remaining. ■

Corollary. No assignment splitting those six points into two groups of three, with all cross-pairs required to avoid w , can exist. Each pair needs both its elements in the same group to avoid being a violating cross-pair; a group of size 3 can hold at most one pair intact (using two of its three slots); two groups can jointly hold at most two of the three pairs intact; the third pair is unavoidably split, and a split pair is exactly a cross-pair landing on w . ■

This directly rules out Attempt 2 for every choice of w , not merely the one tested.

3. A valid 4-level shape exists

An exhaustive search over grade-assignment functions $\{1, \dots, 7\} \rightarrow \{1, \dots, 7\}$, filtered to those using at least four distinct values, found valid examples immediately. A representative: groups $A = \{1, 2\}$, $B = \{3, 4\}$, $C = \{5, 6\}$ (grades 1, 2, 3) and $D = \{7\}$ (grade 4) — sizes $2+2+2+1$. All 21 pairwise products were checked directly against the Weimar-Woods condition and found valid without exception. Critically, $[e_3, e_4] \rightarrow e_7$: B 's self-bracket (grade $2+2=4$) lands exactly at D 's grade 4 — the condition that could never be satisfied in the $1+3+3$ shape now genuinely holds.

4. The four-way split

All 420 valid (partition, grade-assignment) combinations sharing this $2+2+2+1$ size profile were enumerated, and for each, whether the grade-1 and grade-2 groups are self-closed, and whether their self-brackets survive contraction (land exactly at $2 \times$ their own grade), was checked directly.

grade-1 closed	grade-1 survives	grade-2 closed	grade-2 survives	count
No	Yes	No	Yes	84
No	Yes	Yes	No	126
Yes	No	No	No	126
Yes	No	No	Yes	84

Row 2 reproduces the companion paper's original pattern exactly (lower grade open and genuine, higher grade closed and trivial). Row 4 is its mirror: the *lower* grade closed and trivial, the *higher* grade open and genuine — a direct reversal, impossible in the $1+3+3$ shape by construction. Row 1 shows both self-brackets genuinely surviving *simultaneously*, something the earlier shape could never produce regardless of any grouping choice, since grade 4 simply did not exist. Row 3 shows that openness alone is not sufficient — a bracket can exit its own span without landing at the specific grade needed to survive contraction.

5. Why the earlier asymmetry looked like a law

In the $1+3+3$ shape, $\{P, P\}$'s target grade (4) exceeded the shape's maximum (3) unconditionally — closure or not, that bracket was always going to vanish. The observed asymmetry was not evidence that momentum-type closure is *inherently* harmless while coordinate-type closure is *inherently* costly; it was evidence that one side had a real target to protect and the other side's fate was already sealed by the shape before any grouping choice was made. Once a shape provides a real target on both sides — as $2+2+2+1$ does — the asymmetry dissolves into the genuine four-way structure of §4, and either side can play either role.

6. Scope

The mechanism behind Attempt 1's impossibility (§2) was verified computationally but not reduced to a general proof the way Attempt 2's was; whether a comparable Fano-theoretic argument exists is open. The exact counts in §4 (84 vs. 126) sum consistently ($210 + 210 = 420$, an even split by whether the grade-1 group is closed) but their precise combinatorial origin was not derived here. Whether shapes with five or more grade levels produce further, qualitatively new regimes beyond this four-way split is untested.

References

- Companion note, "Which Contractions of the Octonions Give Genuine R-Flux: A Necessary and Sufficient Condition."
- Günaydin, M., Lüst, D. & Malek, E., "Non-associativity in non-geometric string and M-theory backgrounds, the algebra of octonions, and missing momentum modes," arXiv:1607.06474.

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